

Apply silver ion (Ag⁺) antimicrobial coating to the shoulder prosthesis components

Executive Technology Brief

Domain: phage_microbio

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented recurring problem 'adverse_event_without_identified_device_' for shoulder prosthesis: 16 MAUDE reports across 8 distinct devices (20150515 to 20260529); root-cause evidence from 2 recall records and 10 event narratives (custody attached in the campaign report). MAUDE limitations: incidence UNKNOWN, causation UNVERIFIED (Art. XXI.5). Existing solutions fail because: Why existing solutions fail, per retrieved evidence: 2 recall root-cause descriptions (FDA enforcement records, custody attached); 10 MAUDE event narratives from this mechanism cluster (mdr_text, custody attached). Statements are the providers' own words, cust Unresolved gap: Across the 3 relevant retrieved attempts (a bounded, non-exhaustive set), none measurably addresses: conditions, device, devices, events, failure, function, injury, intervention, introducing, maintain, matching, modes, must, problem, producing, prosthesis, rec

Mechanism: Antimicrobial coating of implants with silver to reduce periprosthetic joint infection

Architecture: PROPOSED architecture (no physical system exists): phage loading / immobilization layer; release or contact-kill mechanism; surface substrate (coating architecture); stability preservation element — realizing: Apply silver ion (Ag⁺) antimicrobial coating to the shoulder prosthesis components

Governing Model: Microbiology / phage anti-biofilm governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Key Equations:

BIO-002: Exponential population dynamics. Expression: $N = N_0 * \exp(r * t)$. Applicability to this invention: CONDITIONAL; holds under: unlimited substrate; constant net rate over the window. Assumptions: kill kinetics on an immobilized surface are not well-mixed growth — re-measure before quantitative use. Engaged variables: NOT ESTABLISHED.

BIO-003: Surface-limited adsorption kinetics. Expression: $dN_{\text{infected}}/dt = k_{\text{ads}} * N_{\text{phage}} * N_{\text{bacteria}}$. Applicability to this invention: APPLICABLE; holds under: well-mixed suspension; constant k_{ads} . Assumptions: surface-immobilized phage dosing is not well-mixed — use contact-area density instead (MODEL_DERIVED adaptation); k_{ads} is strain- and environment-specific. Engaged variables: dN_{infected}/dt .

BIO-001: Multiplicity of infection. Expression: $MOI = N_{\text{phage}} / N_{\text{bacteria}}$. Applicability to this invention: APPLICABLE; holds under: well-mixed system; planktonic or defined-surface assay. Assumptions: surface-immobilized phage dosing is not a well-mixed MOI — use contact-area density instead (MODEL_DERIVED adaptation). Engaged variables: MOI.

Kill Condition: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

Current State: ENGINEERING_DEFINITION. No physical prototype. No clinical validation.

Disclosure: This dossier distinguishes established evidence from proposals and unresolved questions.